

# Build & Deployment Notes — Project Pegasus Sprint 3

| Field       | Value                     |
|-------------|---------------------------|
| Date        | 2026-04-09                |
| Sprint      | Sprint 3 (Mar 31 — Apr 9) |
| Prepared by | QA Team                   |
| Status      | In Progress               |

## 1. Docker Images

A total of **16 Docker images** were built and pushed to `docker.io/jnrjose/` during Sprint 3.

| #  | Image Name                         | Registry  |
|----|------------------------------------|-----------|
| 1  | jnrjose/ubn-partner-gateway:latest | docker.io |
| 2  | jnrjose/ubn-admin-gateway:latest   | docker.io |
| 3  | jnrjose/ubn-user-management:latest | docker.io |
| 4  | jnrjose/ubn-admin-auth:latest      | docker.io |
| 5  | jnrjose/ubn-billing:latest         | docker.io |
| 6  | jnrjose/ubn-wallet:latest          | docker.io |
| 7  | jnrjose/ubn-customer:latest        | docker.io |
| 8  | jnrjose/ubn-webhook:latest         | docker.io |
| 9  | jnrjose/ubn-kyc:latest             | docker.io |
| 10 | jnrjose/ubn-account:latest         | docker.io |
| 11 | jnrjose/ubn-payment:latest         | docker.io |
| 12 | jnrjose/ubn-collection:latest      | docker.io |
| 13 | jnrjose/ubn-bank-adapter:latest    | docker.io |
| 14 | jnrjose/ubn-notification:latest    | docker.io |
| 15 | jnrjose/ubn-partner-portal:latest  | docker.io |
| 16 | jnrjose/ubn-admin-portal:latest    | docker.io |

## 2. Deployment Targets

### Migration: Cloudflare Containers to VPS (EC2)

During Sprint 3 the deployment target was **migrated from Cloudflare Containers to a VPS hosted on AWS EC2**. The primary drivers for this migration were:

- Greater control over networking and port mapping.
- Simplified debugging via SSH access to the host.
- Reduced cold-start latency compared to container-based edge deployments.

- Ability to run long-lived background services (Kafka consumers, scheduled jobs).

The VPS runs Docker Compose for orchestration in the dev environment. Production will use a similar EC2-based setup with managed database backends.

### 3. CI/CD Pipeline

**Platform:** GitHub Actions

The pipeline follows a linear stage progression:



Pipeline Stages

| Stage              | Description                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| SAST               | Static Application Security Testing — scans source code for known vulnerability patterns and secrets before build.               |
| Docker Build       | Builds the service image using the Dockerfile in each service repository. Multi-stage builds are used to keep final images lean. |
| Push to Docker Hub | Tags the image as latest and pushes to docker.io/jnrjose/.                                                                       |
| Deploy to VPS      | SSH into the EC2 instance, pulls the latest image, and restarts the container via Docker Compose.                                |
| Slack Notification | Posts deployment status (success/failure) to the #pegasus-deployments Slack channel.                                             |

### 4. Startup Sequence

Services must be started in a defined order to satisfy dependency requirements. The startup sequence is organized into six tiers:

| Tier   | Category             | Services                                                                   |
|--------|----------------------|----------------------------------------------------------------------------|
| Tier 0 | Infrastructure       | Redis, SQL Server, Kafka                                                   |
| Tier 1 | Leaf Services        | Notification, Webhook                                                      |
| Tier 2 | Core Services        | Admin Auth, User Management                                                |
| Tier 3 | Domain APIs          | Account, Payment, Collection, KYC, Billing, Wallet, Customer, Bank Adapter |
| Tier 4 | Event Consumers      | Payment consumer, Collection consumer, Webhook dispatcher                  |
| Tier 5 | Frontends & Gateways | Partner Gateway, Admin Gateway, Partner Portal, Admin Portal               |

Services at each tier depend on all tiers below them being healthy before they can start successfully.

### 5. Changes This Sprint

Commit Activity

| Contributor | CI/CD Commits |
|-------------|---------------|
| Chijindu    | 190           |

|              |            |
|--------------|------------|
| Ezekiel-byte | 55         |
| <b>Total</b> | <b>245</b> |

## Notable Changes

- **Slack Notifications** — Added post-deployment Slack notifications to `#pegasus-deployments` for every service. Notifications include service name, image tag, environment, and deployment status.
- **CODEOWNERS Files** — Added `CODEOWNERS` files to all service repositories to enforce required code reviews. Each service is owned by its respective team lead.
- **Secrets Stripped from appsettings.json** — All connection strings, API keys, and credentials were removed from checked-in `appsettings.json` files and migrated to environment variables / GitHub Secrets. This addresses a security finding from Sprint 2.
- **Swagger Enabled in Staging** — Swagger UI is now enabled for all API services when running in the `Staging` environment. Previously it was only available in `Development`.

## 6. Dev Environment URLs

| #  | Service         | URL                                                                                         | Port |
|----|-----------------|---------------------------------------------------------------------------------------------|------|
| 1  | Partner Gateway | <a href="https://dev-partner-gw.onecluster.co">https://dev-partner-gw.onecluster.co</a>     | 5100 |
| 2  | Admin Gateway   | <a href="https://dev-admin-gw.onecluster.co">https://dev-admin-gw.onecluster.co</a>         | 5200 |
| 3  | User Management | <a href="https://dev-user-mgmt.onecluster.co">https://dev-user-mgmt.onecluster.co</a>       | 5010 |
| 4  | Admin Auth      | <a href="https://dev-admin-auth.onecluster.co">https://dev-admin-auth.onecluster.co</a>     | 5020 |
| 5  | Billing         | <a href="https://dev-billing.onecluster.co">https://dev-billing.onecluster.co</a>           | 5030 |
| 6  | Wallet          | <a href="https://dev-wallet.onecluster.co">https://dev-wallet.onecluster.co</a>             | 5040 |
| 7  | Customer        | <a href="https://dev-customer.onecluster.co">https://dev-customer.onecluster.co</a>         | 5050 |
| 8  | Webhook         | <a href="https://dev-webhook.onecluster.co">https://dev-webhook.onecluster.co</a>           | 5060 |
| 9  | KYC             | <a href="https://dev-kyc.onecluster.co">https://dev-kyc.onecluster.co</a>                   | 5070 |
| 10 | Account         | <a href="https://dev-account.onecluster.co">https://dev-account.onecluster.co</a>           | 5080 |
| 11 | Payment         | <a href="https://dev-payment.onecluster.co">https://dev-payment.onecluster.co</a>           | 5090 |
| 12 | Collection      | <a href="https://dev-collection.onecluster.co">https://dev-collection.onecluster.co</a>     | 5110 |
| 13 | Bank Adapter    | <a href="https://dev-bank-adapter.onecluster.co">https://dev-bank-adapter.onecluster.co</a> | 5120 |
| 14 | Notification    | <a href="https://dev-notification.onecluster.co">https://dev-notification.onecluster.co</a> | 5130 |
| 15 | Partner Portal  | <a href="https://dev-partner.onecluster.co">https://dev-partner.onecluster.co</a>           | 3000 |
| 16 | Admin Portal    | <a href="https://dev-admin.onecluster.co">https://dev-admin.onecluster.co</a>               | 3001 |

## 7. Known Deployment Issues

### 7.1 Partner Gateway — HTTP 502 Bad Gateway

- **Symptom:** Requests to the Partner Gateway return `502 Bad Gateway` intermittently.
- **Impact:** Partner API consumers cannot reach backend services through the gateway.

- **Root Cause (suspected):** The gateway is unable to route to downstream services that have not yet started or are unreachable on the internal Docker network.
- **Status:** Under investigation.

## 7.2 Nine Banking Services Unreachable

- **Symptom:** Nine of the backend services behind the Partner Gateway are not responding to health checks.
- **Services Affected:** Account, Payment, Collection, KYC, Billing, Wallet, Customer, Bank Adapter, Notification.
- **Impact:** All domain API functionality is unavailable in the dev environment.
- **Root Cause (suspected):** Services are either not starting due to missing environment variables or database connectivity issues on the VPS.
- **Status:** Under investigation. Requires SSH access to the VPS to inspect container logs.

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## 8. Rollback Procedure

### Kubernetes (if applicable)

```
kubectl rollout undo deployment/<service-name> -n pegasus
```

### Docker Compose (current dev setup)

```
# Stop the current container
docker stop <container-name>

# Pull the previous image tag
docker pull jnrjose/<service-name>:<previous-tag>

# Restart with the previous tag
docker compose up -d <service-name>
```

Rollback should be performed within **15 minutes** of a failed deployment. If the previous tag is unknown, check the Docker Hub repository tags or the GitHub Actions workflow run history.

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## 9. Database Migrations Applied

The following Entity Framework Core migrations were applied during Sprint 3:

| Migration Name               | Service                    | Description                                                                                                    |
|------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------|
| RenameAdminPermissionToOwner | Admin Auth                 | Renames the Admin permission/role to Owner to better reflect the access level semantics across the platform.   |
| AddKycCompleted              | User Management / Customer | Adds a KycCompleted boolean column to the partner/customer record, used to gate access to production API keys. |

### Applying Migrations

Migrations are applied automatically on service startup via `Database.Migrate()` in the `Program.cs` startup pipeline. Manual application can be done with:

```
dotnet ef database update --project <ServiceProject> --startup-project <ServiceProject>
```

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## 10. Summary

Sprint 3 focused heavily on CI/CD maturity and infrastructure migration. The pipeline is functional end-to-end (SAST through Slack notification), and all 16 Docker images build and push successfully. The primary blockers are the VPS networking issues causing 9 services to be unreachable and the Partner Gateway 502 errors. These must be resolved before Sprint 4 testing can proceed effectively.